

Zhuohan Wang

zhuohan_wang@g.harvard.edu | slarkprime@gmail.com | Google Scholar

Education

Harvard University

M.S. in Computational Science and Engineering

Cambridge, MA

Sep 2025 – Present

City University of Hong Kong

BEng in Computer and Data Engineering

Hong Kong

Aug 2021 – Jun 2025

- **GPA: 3.95** (Rank 1st / 69); **Summa Cum Laude**; HKSAR Government Scholarship

Professional Experience

Mitsubishi Electric Research Laboratories (MERL)

Research Intern; Mentors: Dr. Pu (Perry) Wang, Dr. Anoop Cherian

Cambridge, MA

May 2026 – Sep 2026

Publications: 3 papers under review at ICLR 2027, 2 as first author

- Tested whether **vision-language models** can use the image to correct false visual claims in their own reasoning, designing a controlled replay evaluation which shows that they largely fail to, even with the evidence still in context.
- Found that copying RGB chain-of-thought into **radar and audio** students hurts, and designed a method that re-derives each rationale against the target sensor first, beating naive distillation on every backbone tested.
- Defined what a benchmark for evidence-grounded sensor reasoning about human activity should measure, separating **understanding** from **reasoning**; designed the question types and built the question-generation pipeline, then handed it off to the data-collection and evaluation teams.

SenseTime Research

Algorithm Researcher Intern; Advisor: Dr. Litong Feng

Hong Kong

Jul 2024 – Sep 2025

- **Improving Instruction-Following in Proprietary LLM:** Participated in data workflows and optimization for **SenseChat**, contributing to **SFT**, **DPO**, and **RL** stages. Took independent responsibility for improving **instruction-following ability**, achieving a **top-five ranking** on the **SuperCLUE** benchmark (bi-monthly updates), and driving additional gains on **LiveBench** and **IFEval** compared to leading models such as Claude and OpenAI's latest releases.
- **Dataset and benchmark construction:** Contributed to the design and construction of datasets and benchmarks for tasks such as **safety**, **Sudoku**, and **role-playing**, achieving consistent gains of **2–3 percentage points** across multiple evaluation tracks.
- Engaged in **multi-agent research** aimed at enhancing **hard mathematical reasoning** capabilities of LLMs.

SHANGHAI-FANUC Robotics

Machine Vision Engineer Intern

Shanghai, China

Jul 2022 – Aug 2022

- **Calibration and segmentation accuracy:** Improved calibration and segmentation accuracy by developing and refining computer-vision algorithms.
- **Visualization tools for decision support:** Integrated tailored visualization tools for calibration and segmentation, aiding decision-making and stakeholder communication.

Research Experience

LLM Reasoning Evaluation and Efficient Decoding

Independent Researcher, Harvard University

Cambridge, MA

Sep 2025 – Present

Publications: 2 accepted at NeurIPS 2026 (1 Spotlight, 1 oral nomination) [1][3]; 2 under review at ICLR 2027

- Led **OracleLadder** to locate where LLM math reasoning breaks, giving models increasing **oracle help** and grading answers with a symbolic verifier; in six models (8B–671B), the **composition gap** (every step solved, whole problem failed) was the largest failure (**33–48%** of problems).
- Tested how well **process reward models**, validated as step checkers, choose branches in test-time search by scoring each choice on final correctness, showing that step correctness is a weak signal for branch selection.
- Helped formulate the idea of **D-PACE**, a **speculative decoding** drafter loss that weights each position by its contribution to accepted length; then, as co-first author, extended it to weight positions by the true acceptance probability.

Causal Sufficiency and Necessity for Chain-of-Thought Reasoning

Research Assistant; Advisor: Dr. Mengyue Yang

London, UK

Jan 2025 – Jul 2025

Publications: 1 accepted at NeurIPS 2025, co-first author [2]

- Introduced a **causal framework** for **Chain-of-Thought** reasoning (**sufficiency & necessity**) to rigorously identify **essential reasoning steps**.
- Developed a **PNS-guided pruning** method that **improves accuracy** on easier benchmarks while **significantly reducing**

token usage, and on harder benchmarks achieves **20–60% fewer tokens/steps** while maintaining **comparable accuracy**.

Parameter-Efficient Fine-Tuning for Large Language Models

Research Assistant; Advisor: Prof. Lai-Man Po

Hong Kong

Jun 2023 – Mar 2024

Publications: 1 accepted at ICONIP 2024 [4]

- Conducted research on **parameter-efficient fine-tuning (PEFT)** for LLMs, developing **LoRA** and **GaLore** variants that **improved convergence, reduced memory footprint and training cost**, and **enhanced benchmark performance** across multiple tasks, including **SBoRA**.
- Proposed and developed **Colore**, a novel PEFT method for LLMs achieving **2× training speed** over GaLore under specific settings, while maintaining **competitive benchmark performance**.

Large-Scale Entity Matching and Embedding-based Record Linkage

Research Engineer; Advisor: Prof. Rengong Zhang

Hong Kong

Jun 2023 – Jul 2024

- Contributed to a **government-funded research project** on large-scale **entity matching of company names**. Built a system operating on **terabyte-scale data**, integrating heterogeneous databases and leveraging **embedding models** to deliver an **automated, high-accuracy pipeline**.
- Developed a **data pipeline** for an **HR consulting project** that mapped project names to ranking categories. Applied **clustering techniques** and LLMs to improve **matching precision** and deliver a **scalable automated solution**.

Selected Publications (* equal contribution)

Reasoning

[1] **Solving Every Step Is Not Enough: Milestone Oracles Reveal a Composition Gap in LLM Math Reasoning**

Z. Wang (project lead), H. Ma, T. Wu, Y. Duan, Z. Liao, J. Yu

NeurIPS 2026 (Oral nomination). [arXiv] [Paper] [Code]

[2] **Causal Sufficiency and Necessity Improves Chain-of-Thought Reasoning**

X. Yu*, Z. Wang*, L. Yang, H. Li, A. Liu, X. Xue, J. Wang, M. Yang

NeurIPS 2025. [arXiv] [Code]

Efficient Training and Inference

[3] **D-PACE: Dynamic Position-Aware Cross-Entropy for Parallel Speculative Drafting**

T. Wu, Y. Yao, Z. Qi, H. Zheng, C. Zhang, Z. Wang, H. Ma, Z. Liao, H. Lakkaraju, J. Li, Y. Du

NeurIPS 2026 (Spotlight). [Paper] [Code]

[4] **SBoRA: Low-Rank Adaptation with Regional Weight Updates**

L.-M. Po, Y. Liu, H. Wu, T. Zhang, W.-Y. Yu, Z. Wang, Z. Jiang, K. Li

ICONIP 2024. [arXiv] [Code]

Skills & Languages

Programming: Python (primary), C++, Java, SQL, R, MATLAB, Assembly

ML/AI: PyTorch, scikit-learn; SGLang, vLLM; OpenAI APIs; **Cloud/DevOps:** AWS, Docker, Git

Languages: Native Mandarin & Shanghaiese; Proficient English **Interests:** Badminton, fitness, table tennis, jogging, investment, emerging technologies, listening to podcasts